

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Data Interpretation

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 1 – Data Interpretation
Weightage:	35% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	AI & ML	Date:	19-08-26

Code of Conduct

I A. Satya Manoj 192425370 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Satya Manoj

Instructions

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems.	Q1
First-Order Predicate Calculus & Representing Linguistically Relevant Concepts	Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems.	Q3
Syntax-Driven Semantic Analysis	Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications.	Q4

1. Frame Semantics in Banking Customer-Service Systems

A digital banking platform uses semantic frames to understand customer requests. The NLP system identifies the **action, agent, object, and destination** involved in each request.

Customer Query	Generated Semantic Frame
"Transfer ₹20,000 from my savings account to my son's account."	TRANSFER(SourceAccount, Amount, DestinationAccount, Customer)
"Block my debit card immediately."	BLOCK(DebitCard, Customer)
"Send my transaction statement to my email."	SEND(TransactionStatement, Email, Customer)
"Increase my daily withdrawal limit."	MODIFY(WithdrawalLimit, Customer)

Additional transaction logs show:

Query ID	Actual User Intent	System Interpretation
B1	Transfer money to another account	Transfer money
B2	Block debit card	Block debit card
B3	Receive transaction statement by email	Send statement to email
B4	Increase withdrawal limit	Decrease withdrawal limit

Tasks:

1. Analyze the semantic frame of each banking request and identify the relationship between the action and its participants.
2. Identify the query in which the semantic interpretation is incorrect and explain why.
3. Analyze how incorrect identification of semantic roles can lead to an incorrect banking operation.
4. Recommend a frame-based semantic analysis approach to improve the reliability of banking conversational systems.

2. First-Order Predicate Calculus for Smart Agriculture

A smart agricultural system monitors crop fields using sensors and logical rules.

Field Data:

Field	Soil Condition	Irrigation Status	Crop
F1	Dry	Available	Rice
F2	Wet	Available	Wheat
F3	Dry	Unavailable	Maize
F4	Wet	Unavailable	Rice

The system uses the following predicate rules:

- $Dry(x) \wedge IrrigationAvailable(x) \rightarrow NeedsWater(x)$
- $NeedsWater(x) \rightarrow Irrigate(x)$
- $Wet(x) \rightarrow \neg NeedsWater(x)$
- $IrrigationUnavailable(x) \rightarrow \neg Irrigate(x)$
- $Rice(x) \wedge NeedsWater(x) \rightarrow PriorityIrrigation(x)$

Tasks:

1. Represent the field observations using First-Order Predicate Calculus.
2. Using the given rules, determine which fields require irrigation.
3. Determine whether F1 and F3 should be treated differently by the automated irrigation controller.
4. Analyze whether predicate logic alone is sufficient for agricultural decision support when sensor information is incomplete or uncertain.

3. Word Sense Disambiguation in a Travel Recommendation System

A travel platform receives search queries containing ambiguous words. The recommendation engine uses surrounding words and user interactions to determine the intended meaning.

Search Query	Possible Sense 1	Possible Sense 2
"Amazon tour"	Amazon rainforest	Amazon company
"Safari booking"	Wildlife tour	Web browser
"Java vacation"	Java island	Java programming language
"Cruise package"	Ship journey	Cruise software/project

The system records the following user interactions:

Query	User Interaction
Amazon tour	Viewed rainforest trekking packages
Safari booking	Selected wildlife resort packages
Java vacation	Viewed hotels in Bali and Java
Cruise package	Viewed Mediterranean ship packages

However, another user searches: **"Safari download for Windows"** and clicks a browser download page.

Tasks:

1. Determine the intended sense of the ambiguous terms using the available context.
2. Identify the lexical and contextual cues that distinguish the possible senses.
3. Analyze why the same word may require different interpretations for different users.
4. Design an industrial-scale WSD strategy using query context, user history, embeddings, and click behaviour to improve travel recommendations.

4. Syntax-Driven Semantic Analysis in Legal Document Processing

A legal NLP system analyzes sentences from contracts and assigns semantic roles based on their syntactic structures.

The system produces the following analyses:

Legal Sentence	Parse Structure	Generated Semantic Interpretation
"The supplier delivered the equipment to the buyer."	Subject–Verb–Object	Supplier = Agent, Equipment = Object, Buyer = Recipient
"The buyer received the equipment from the supplier."	Subject–Verb–Object	Buyer = Agent, Equipment = Object, Supplier = Recipient
"The company terminated the contract after the violation."	Subject–Verb–Object	Company = Agent, Contract = Object
"The contract was terminated by the company."	Passive Construction	Contract = Agent, Company = Object

Tasks:

1. Analyze how syntactic structure, including active and passive constructions, influences semantic-role assignment.
2. Identify the incorrect semantic-role assignments in the given interpretations.
3. Explain how confusing grammatical subject with semantic agent can affect automated legal information extraction.
4. Recommend a syntax-driven semantic analysis architecture that can correctly handle active voice, passive voice, prepositional phrases, and long-distance dependencies in legal documents.

1. Frame Semantics in Banking Customer-Service Systems

1. Analyze the semantic frame of each banking request and identify the relationship between the action and its participants.

Frame semantics identifies the action and its participants in a sentence.

Query 1: "Transfer ₹20,000 from my savings account to my son's account."

- Action: Transfer
- Source: Savings account
- Amount: ₹20,000
- Destination: Son's account
- Customer: Person requesting the transfer

Query 2: "Block my debit card immediately."

- Action: Block
- Object: Debit card
- Customer: Person requesting the action

Query 3: "Send my transaction statement to my email."

- Action: Send
- Object: Transaction statement
- Destination: Email
- Customer: Person requesting it

Query 4: "Increase my daily withdrawal limit."

- Action: Modify
- Object: Withdrawal limit
- Modification: Increase
- Customer: Person requesting it

2. Identify the query in which the semantic interpretation is incorrect and explain why.

B4 is incorrect. The actual intent is to increase the withdrawal limit, but the system interprets it as decrease.

Correct interpretation:

MODIFY(WithdrawalLimit, Increase, Customer)

The system identified the correct object but reversed the intended action.

3. Analyze how incorrect identification of semantic roles can lead to an incorrect banking operation.

Incorrect semantic roles can cause serious banking errors. For example, confusing the source and destination accounts may transfer money to the wrong account. Similarly, interpreting "increase" as "decrease" may reduce the customer's withdrawal limit. Such errors can cause financial loss, security problems, and customer dissatisfaction.

4. Recommend a frame-based semantic analysis approach.

A reliable banking system should:

1. Detect the user's intent.
2. Identify semantic roles such as amount, source, destination, and object.
3. Construct a semantic frame.
4. Resolve entities using context.
5. Validate the extracted information.
6. Confirm sensitive transactions with the customer.

This approach improves the accuracy and safety of banking conversational systems.

2. First-Order Predicate Calculus for Smart Agriculture

1. Represent the field observations using First-Order Predicate Calculus.

The observations can be represented as:

- **F1:** Dry(F1), IrrigationAvailable(F1), Rice(F1)
- **F2:** Wet(F2), IrrigationAvailable(F2), Wheat(F2)
- **F3:** Dry(F3), IrrigationUnavailable(F3), Maize(F3)
- **F4:** Wet(F4), IrrigationUnavailable(F4), Rice(F4)

Given rules:

- $\text{Dry}(x) \wedge \text{IrrigationAvailable}(x) \rightarrow \text{NeedsWater}(x)$
- $\text{NeedsWater}(x) \rightarrow \text{Irrigate}(x)$
- $\text{Wet}(x) \rightarrow \neg \text{NeedsWater}(x)$
- $\text{IrrigationUnavailable}(x) \rightarrow \neg \text{Irrigate}(x)$
- $\text{Rice}(x) \wedge \text{NeedsWater}(x) \rightarrow \text{PriorityIrrigation}(x)$

2. Using the given rules, determine which fields require irrigation.

F1: F1 is dry and irrigation is available.

$\text{Dry}(F1) \wedge \text{IrrigationAvailable}(F1) \rightarrow \text{NeedsWater}(F1)$

Therefore, $\text{Irrigate}(F1)$.

Since it is rice, $\text{PriorityIrrigation}(F1)$ also follows.

F2: F2 is wet.

$\text{Wet}(F2) \rightarrow \neg \text{NeedsWater}(F2)$

Therefore, F2 does not require irrigation.

F3: F3 is dry, but irrigation is unavailable.

$\text{IrrigationUnavailable}(F3) \rightarrow \neg \text{Irrigate}(F3)$

Therefore, F3 cannot be irrigated.

F4: F4 is wet and irrigation is unavailable.

$\text{Wet}(F4) \rightarrow \neg \text{NeedsWater}(F4)$

Therefore, F4 does not require irrigation.

3. Determine whether F1 and F3 should be treated differently.

Yes. Both fields are dry, but irrigation availability is different.

- **F1:** Dry + irrigation available → Irrigate
- **F3:** Dry + irrigation unavailable → Do not irrigate

Therefore, the controller should irrigate F1 but not F3.

4. Analyze whether predicate logic alone is sufficient.

Predicate logic is useful for clear rule-based decisions, but it is not sufficient when sensor information is incomplete or uncertain. Sensors may produce noisy or inaccurate values.

Therefore, practical agricultural systems can also use fuzzy logic, probability, machine learning, and sensor confidence values.

Thus, predicate logic should be combined with other techniques for reliable agricultural decision-making.

3. Word Sense Disambiguation in a Travel Recommendation System

1. Determine the intended sense of the ambiguous terms.

- **Amazon tour** → **Amazon rainforest**
The user viewed rainforest trekking packages.
- **Safari booking** → **Wildlife tour**
The user selected wildlife resort packages.
- **Java vacation** → **Java island**
The user viewed hotels in Bali and Java.
- **Cruise package** → **Ship journey**
The user viewed Mediterranean ship packages.
- **Safari download for Windows** → **Web browser**
The words "download" and "Windows" indicate software.

2. Identify the lexical and contextual cues.

The system uses surrounding words and user behaviour.

- **Amazon + tour/trekking** → Rainforest
- **Safari + booking/wildlife** → Wildlife tour
- **Java + vacation/hotels/Bali** → Java island
- **Cruise + Mediterranean/ship** → Ship journey
- **Safari + download/Windows** → Web browser

User history and clicked pages also provide useful contextual information.

3. Analyze why the same word may require different interpretations.

The meaning of a word depends on the user's context and intention.

For example:

- **"Safari booking"** → Wildlife safari

- **"Safari download for Windows"** → Web browser

Therefore, WSD must consider the query, surrounding words, user history, and user behaviour rather than relying only on the word itself.

4. Design an industrial-scale WSD strategy.

A travel recommendation system can use:

1. Query preprocessing to identify ambiguous words.
2. Context analysis to examine surrounding words.
3. User history such as searches and bookings.
4. Embeddings to understand semantic meaning.
5. Click behaviour as user feedback.
6. Sense scoring to select the most likely meaning.
7. Continuous learning from user interactions.

For example:

Java + vacation + hotels → Java island

while:

Java + programming + tutorial → Java programming language

This improves the accuracy of travel recommendations.

4. Syntax-Driven Semantic Analysis in Legal Document Processing

1. Analyze how syntactic structure influences semantic-role assignment.

Syntactic structure helps identify the semantic roles of entities. However, the grammatical subject is not always the semantic agent.

"The supplier delivered the equipment to the buyer."

- Supplier = Agent
- Equipment = Object/Theme
- Buyer = Recipient

"The contract was terminated by the company."

- Contract = Patient/Theme
- Company = Agent

Therefore, active and passive constructions must be analyzed differently.

2. Identify the incorrect semantic-role assignments.

Sentence: "The buyer received the equipment from the supplier."

The system says:

- Buyer = Agent
- Supplier = Recipient

This is incorrect.

Correct roles:

- Buyer = Recipient
- Equipment = Object/Theme
- Supplier = Source

Sentence: "The contract was terminated by the company."

The system says:

- Contract = Agent
- Company = Object

This is incorrect.

Correct roles:

- Contract = Patient/Theme
- Company = Agent

3. Explain how confusing grammatical subject with semantic agent can affect legal information extraction.

A grammatical subject is not always the person performing the action.

In:

"The contract was terminated by the company."

the subject is **contract**, but the semantic agent is **company**.

If the system confuses these roles, it may assign responsibility to the wrong entity. This can affect legal summaries, contract analysis, liability identification, and information extraction.

4. Recommend a syntax-driven semantic analysis architecture.

A suitable architecture is:

Legal Text → Preprocessing → NER → Dependency Parsing → Voice Detection → Semantic Role Labeling → Prepositional Phrase Analysis → Coreference Resolution → Legal Validation

Important considerations are:

- Active voice: Subject often acts as Agent.
- Passive voice: Agent may occur after **"by"**.
- **"to"** can indicate a Recipient.
- **"from"** can indicate a Source.
- Coreference resolution helps connect entities across sentences.

This architecture improves semantic-role identification in complex legal documents.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria Level	Understanding of Data	Analysis	Application of Concepts	Conclusion	Clarity	Sub. Total	Total %
1	3	3	3	3	4		76
2	3	3	3	3	4		
3	3	3	3	3	4		
4	3	3	3	3	4		

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